

Debjcet Mazumder

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Debjcet-Mazumder

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About Me

Driven B.Tech student and full-stack developer with a track record of architecting memory-optimized mobile apps, interactive WebGL interfaces, and hardware-integrated IoT systems. Passionate about continuous learning, I focus on strengthening programming fundamentals by building and iterating on complex, real-world solutions.

Skills & Achievements

- Languages:** Python, Java, C/C++, HTML5, CSS3
- Applied Experience & Tools:** JavaScript, Kotlin, Dart, Git, GitHub
- Frameworks & Libraries:** Flutter, Flask, GSAP, Jinja2
- Core Concepts & Databases:** Object-Oriented Programming (OOP), Data Structures & Algorithms, MySQL, SQLite, SQLAlchemy
- Design & Modeling:** AutoCAD, Blender, Adobe Photoshop
- Achievements:** JISTech 2K26: 1st place - App-Dev Competition, 2nd Runner-Up - Theme Based Project (AES)

Projects

Maargdarshan | AI-Enabled Smart Navigation Ecosystem  *C/C++, Flutter, Kotlin, Python, JavaScript, Dart, ESP32, Arduino*
[Team Project - Founder, Solo Architect & Lead Developer]

Architecture: Three-Tier Ecosystem | Website  | App  (*Prototypes in active Beta testing)

- Hardware Node:** Engineered a custom wearable IoT device integrating a multidirectional spatial awareness array for real-time environmental mapping and localized haptic response.
- Web Backend:** Architected a low-latency web portal featuring a digital twin interface to visually track hardware orientation, integrated with real-time optical telemetry, historical spatial-analysis logs alongside emergency dispatch protocol.
- Frontend & UI Engineering:** Engineered a high-performance, visually thematic UI utilizing advanced rendering techniques to achieve fluid physics (satin cloth and water ripple effects). Integrated dynamic thematic motifs (peacock feather cursor) to visually translate the project's assistive philosophy into an interactive digital experience.
- Mobile Application:** Developed a memory-optimized native app in a headless ambient state, seamlessly unifying voice-navigated spatial AI, optical telemetry, and spatial-analysis logs with an OS-bypassing emergency routing protocol.
- Backend Routing Server:** Architected the central backend API to bridge the entire ecosystem, handling real-time data flow, sensor telemetry routing, SOS dispatch routing, live optical telemetry streaming with low latency and state synchronization between the IoT hardware, mobile application, and web portal.

Super Wheels | High-Performance Automotive UI/UX   [Solo Project] *HTML5, CSS3, JS, GSAP*

- Developed a high-fidelity interactive landing page prototype utilizing native web technologies without relying on heavy frontend frameworks.
- Designed complex GSAP scroll-triggered animations, physics-based CSS hover states, and a custom JavaScript dual-audio controller for synchronized multimedia playback.

Udyog | AI-Driven Industrial Supply & Waste Exchange  *SQLAlchemy, Flask, Python, SQLite*
[Duo Project - Lead Developer & Designer] (Under Development)

- Architected a full-stack B2B marketplace featuring a Circular Economy module, enabling factories to seamlessly purchase raw materials and list manufacturing by-products as secondary resources.
- Engineered an AI recommendation engine utilizing geospatial proximity and sentiment analysis to optimize supply chain discovery, backed by dynamic SQLite queries for complex material filtering.
- Implemented an industrial compliance layer with digital certificate verification and a bespoke multi-tag rating system to establish B2B trust via actionable engineering metrics.

Technical Practice & Implementation

Core Fundamentals & Engineering Practice *Ongoing*

- Building custom implementations of arrays, linked lists, and sorting algorithms to strengthen computational problem-solving.
- Applying Object-Oriented Programming principles to design structured backend data models and writing optimized queries for relational databases.
- Initializing repositories, utilizing Git for structured version tracking, and systematically documenting code progress on GitHub.

Education

B.Tech in Computer Science and Engineering (2025-2029)

JIS College of Engineering, Kalyani

Pursuing

Interests

- Participating in hackathons and tech events
- Full-stack web and mobile development
- Building hardware and sensor-based projects connected to web interfaces